

LOGESH PERUMAL C

AI DEVELOPER

Salem, TN

[LinkedIn](#)

+91 9362003407

[Github](#)

clogeshperumal2004@gmail.com

[Portfolio](#)

PROFESSIONAL SUMMARY •

Results-driven AI Developer with **1.5+ years of experience** building and deploying scalable AI systems across Machine Learning, Deep Learning, NLP, and Generative AI. Proven expertise in end-to-end AI pipelines — from data preprocessing and model development to production deployment via RESTful APIs. Experienced in LLM-powered applications using **LangChain, LangGraph, RAG, OpenAI API, and Azure OpenAI**, with a demonstrated track record of achieving **93.5% model accuracy** through fine-tuning and hyperparameter optimization. Adept at integrating AI into full-stack applications to deliver robust, business-focused solutions.

PROFESSIONAL EXPERIENCE •

Associate AI Developer

June 2025 – Present

FocusR Consultancy and Technologies Pvt Limited

Onsite, India

- Designed and deployed end-to-end AI/ML pipelines covering data ingestion, preprocessing, model training, evaluation, and production deployment using **Python, TensorFlow, and Scikit-Learn**.
- Built **Generative AI and Agentic AI** applications leveraging LLMs (OpenAI API, Azure OpenAI) with **LangChain and LangGraph** for agent workflows, RAG pipelines, and intelligent conversational systems.
- Engineered **prompt optimization frameworks** with structured testing, iterative refinement, and hallucination reduction strategies to improve LLM reliability and output consistency.
- Implemented **MCP (Model Context Protocol)** for structured tool interaction, enabling improved context management and orchestration in LLM-based systems.
- Implemented **OCR/ICR-based document processing** solutions for automated extraction, classification, and validation of structured and unstructured data.
- Fine-tuned and evaluated **NLP models** for text classification, named entity recognition (NER), and sentiment analysis using **Hugging Face Transformers and NLTK**.
- Deployed ML models as **RESTful APIs** using FastAPI and Flask, integrated with React and Node.js to deliver full-stack AI-powered applications.
- Built **vector database pipelines (ChromaDB)** for RAG systems, enhancing semantic search accuracy and knowledge retrieval performance.

INTERNSHIP EXPERIENCE •

Intern Consultant – AI & Data Analytics

Dec 2024 – June 2025

FocusR Consultancy and Technologies Pvt Limited

Onsite, India

- Developed and deployed **full-stack AI-powered applications** using Python, FastAPI, Flask, Django, React, TypeScript, and REST APIs, enabling scalable backend and seamless frontend experiences.
- Worked on **LLM and Generative AI applications** using LangChain, LangGraph, LangSmith, OpenAI API, and Azure OpenAI — implementing RAG pipelines, prompt engineering, and chatbot workflows for intelligent automation.
- Integrated **vector databases (FAISS, ChromaDB, Pinecone)** and MongoDB for efficient data storage, retrieval, and semantic search in RAG-based AI applications.
- Performed **OCR/ICR-based document processing**, data labeling, annotation, and preprocessing using Pandas, NumPy, Matplotlib, and Seaborn to support supervised learning pipelines.

Data Science & ML Intern

Sep 2022

Matt Engineering Solutions

Onsite, India

- Developed and evaluated **ML models** for classification and prediction tasks including Movie Genre Classification, Credit Card Fraud Detection, Customer Churn Prediction, Stock Price Forecasting, Salary Prediction, and Titanic Survival Classification using **Logistic Regression, Random Forest, Decision Trees, SVM, and CNN**.
- Designed and optimized **prompts for LLMs** (OpenAI / Gemini) using few-shot, zero-shot, and chain-of-thought (CoT) techniques — implemented prompt chaining, A/B testing, and hallucination mitigation for chatbots, summarization, and document processing workflows.
- Built a **real-time Face Emotion Detection system** using Haar Cascade Classifier, CNNs, and OpenCV with TensorFlow/Keras — achieving efficient inference with image preprocessing, data augmentation, and normalization.
- Performed **EDA, feature engineering, hyperparameter tuning** (GridSearchCV), and model evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC, R^2 , MAE, and MSE — achieved **92% predictive accuracy** on regression tasks using Python, Pandas, NumPy, Scikit-Learn, Matplotlib, and Seaborn.

TECHNICAL SKILLS •

Languages & Frameworks: Python · FastAPI · Flask · Django · React · Node.js · Next.js · TypeScript · JavaScript · HTML/CSS
AI / ML / Deep Learning: Machine Learning · Deep Learning · ANN · CNN · RNN · TensorFlow · PyTorch · Keras · Scikit-Learn · OpenCV · NLTK · NLP · Computer Vision
Generative AI & LLMs: LangChain · LangGraph · LangSmith · OpenAI API · Azure OpenAI · Hugging Face · RAG · Fine-tuning · Prompt Engineering · Agentic AI · Tool Calling · MCP
Data Science & Analytics: Pandas · NumPy · Matplotlib · Seaborn · EDA · Feature Engineering · Hyperparameter Tuning · Classification · Regression · SQL · Microsoft Excel
Databases & Vector Search: MongoDB · PostgreSQL · ChromaDB · FAISS · Pinecone · Vector Databases · Semantic Search · Embeddings

KEY PROJECTS •

DocuRAG Assistant · *LangChain · LangGraph · LangSmith · OpenAI API · Azure OpenAI · ChromaDB · FastAPI · RAG · Prompt Engineering · MCP · Python*

- Architected an enterprise-grade **Retrieval-Augmented Generation (RAG)** assistant using **LangChain and LangGraph**, integrating OpenAI API and Azure OpenAI with **ChromaDB vector database** for semantic search, context-aware document Q&A, and intelligent knowledge retrieval across unstructured data sources.
- Implemented **agentic AI workflows** using MCP (Model Context Protocol) for structured tool interaction and multi-step reasoning, with LangSmith for pipeline monitoring, tracing, and performance evaluation.
- Engineered **prompt optimization and hallucination mitigation frameworks** with structured A/B testing, iterative refinement, and qualitative/quantitative output evaluation to improve LLM response reliability and factual accuracy.
- Deployed the full system as a **RESTful API using FastAPI**, integrated with a React front-end — delivering an end-to-end, production-ready AI-powered document intelligence application.

NutriVision AI · *CNN · TensorFlow · Keras · Transfer Learning · TensorFlow Hub · OpenCV · Streamlit · PIL · NumPy · Matplotlib · Python*

- Designed and trained two independent **deep learning image classification models** — a Food Classifier and a Vegetable Classifier — using **CNN architecture with Transfer Learning** (TensorFlow Hub pre-trained models), achieving **93.5% classification accuracy** across multiple food and vegetable categories.
- Applied advanced image preprocessing techniques including **data augmentation, normalization, and image resizing** using PIL and OpenCV to improve model generalization and reduce overfitting on unseen data.
- Performed **hyperparameter tuning, model evaluation, and performance benchmarking** using accuracy, precision, recall, and F1-score metrics — leveraging Matplotlib and Seaborn for visual analysis of training/validation loss curves and confusion matrices.
- Deployed both classifiers as **interactive Streamlit web applications** supporting real-time image upload and live camera-based prediction, delivering a production-ready, user-friendly Computer Vision solution.

EDUCATION •

B.Tech — Artificial Intelligence & Data Science Mahendra Engineering College, Namakkal, Tamil Nadu	2021 – 2025 CGPA 9.26 / 10
HSC — Maths & Computer Science Jothi Vidhyalaya Matric Higher Secondary School, Salem	2020 – 2021 88%
SSLC Government Higher Secondary School, Salem	2018 – 2019 82%

CERTIFICATIONS & ACHIEVEMENTS •

◆ IBM Full Stack Software Developer — Coursera	◆ AWS Cloud Practitioner — Coursera
◆ Model Context Protocol — Anthropic	◆ NLP Foundation — Infosys Springboard
◆ Machine Learning A-Z: AI, Python & R — Udemy	◆ ML Foundation — Infosys Springboard
◆ Data Analytics & Visualization — Accenture	◆ Resource Person — AI & ML Workshop (IEC)

PUBLICATION •

Tool Wear Condition Monitoring Using Emitted Sound Signals by Simple Machine Learning Technique
Published · June 2022